
Does Cost-Efficiency Travel? Rank Transfer and Frontier Stability in Fixed-Scaffold Agent Evaluation

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Abstract

1 Agent leaderboards increasingly report task success and dollar cost, inviting prac-
2 titioners to select a “cost-efficient” system from a single benchmark. We test
3 whether this signal transfers across workloads and whether any transfer reflects
4 stable model-label cost propensity. Using public Holistic Agent Leaderboard
5 (HAL) results, we hold the displayed HAL Generalist Agent scaffold fixed and
6 compare repeated displayed model labels across six benchmarks. Raw dollar-cost
7 ranks transfer robustly in several high-overlap pairs: among pairs with at least 12
8 shared labels, cost Spearman correlation ranges from 0.74 to 0.93, survives Holm-
9 Bonferroni correction, and exceeds a label-permutation null. However, a pooled
10 log-cost model with benchmark and displayed-label fixed effects explains 82% of
11 observed log-cost variance; after removing this stable observed label propensity,
12 no high-overlap residual-cost rank association is significant under a 20,000-draw
13 permutation test. Thus raw cost-rank transfer largely encodes persistent label-level
14 cost propensity, which may include price tier, reasoning configuration, token use,
15 routing, and unrecorded setup rather than portable execution dynamics. Cost-
16 accuracy frontier membership remains non-universal: no label tested on at least
17 five workloads lies on every frontier under nondominance, a HAL-inspired convex
18 hull, or a 5% near-tie sensitivity. Public data lack token/provider fields required to
19 isolate historical list pricing.

20 1 Introduction

21 Agent benchmarks evaluate systems that reason over multiple steps, invoke tools, and interact with
22 environments. Reported behavior depends on the language model, scaffold, tools, prompts, execution
23 limits, and task environment. Evaluation is also expensive: HAL reports 21,730 rollouts across nine
24 benchmarks and roughly \$40,000 in cost [Kapoor et al., 2026]. Cost-aware leaderboards therefore
25 plot accuracy against dollar cost and recommend cost-accuracy frontiers.

26 A practical question follows: if a displayed model label is efficient on one agent benchmark, is
27 it a promising low-cost choice elsewhere? This may hold when capability and resource use are
28 portable, but can fail when tool calls, retries, or stopping behavior differ by environment. We study
29 this question descriptively, holding the publicly displayed HAL Generalist Agent scaffold fixed and
30 matching exact public model-label strings across workloads.

31 Our design is coverage-aware. Most HAL scaffolds occur on one benchmark, making a global model-
32 scaffold decomposition unsupported. The Generalist Agent is the only broadly repeated scaffold. We
33 retain six benchmarks with adequate pairwise overlap and analyze accuracy rank, dollar-cost rank,
34 and frontier membership only among shared labels. We contribute: (i) a cross-workload analysis that
35 separates robust high-overlap evidence from low-overlap estimates, (ii) configuration-resampling,
36 Holm-corrected, and label-permutation checks for pairwise ranks, (iii) a cost-propensity adjustment

that tests whether raw cost transfer remains beyond stable displayed-label cost levels, and (iv) a frontier sensitivity analysis separating nondominance, a HAL-inspired hull, and a near-tie tolerance.

2 Related work

Kapoor et al. [2026] introduce HAL as a cost-aware leaderboard across coding, web, science, and customer-service tasks. Kapoor et al. [2024] argue for joint accuracy–cost evaluation of agents. SWE-Effi studies resource-constrained software-engineering agents and reports scaffold–model interaction within one domain [Fan et al., 2025]. Ndzomga [2026] study within-benchmark rank preservation under task reduction, while Benchmark² studies cross-benchmark rank consistency for static LLMs [Qian et al., 2026]. None directly establishes cross-workload rank transfer for interactive-agent dollar costs under a repeated scaffold.

3 Data and method

We analyze the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026]: 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 model labels. Accuracy and total dollar cost are complete, but 235 rows report one run. The frozen source, exact hash, and code are in the anonymous supplement.

The primary cohort fixes the displayed HAL Generalist Agent scaffold: 21 labels on CORE-Bench Hard, 17 on GAIA, 9 on SciCode, 7 on ScienceAgentBench, 18 on SWE-bench Verified Mini, and 14 on TAU-bench Airline. USACO has one Generalist result and is excluded. For each pair with at least five shared labels, we compute Spearman ρ and Kendall rank correlation for accuracy and dollar cost. We use 10,000 configuration-resampling draws and 10,000 label-pairing permutations; these quantify finite-label sensitivity, not rollout-level or population-model uncertainty. Holm–Bonferroni correction is applied separately to 15 accuracy and 15 cost tests.

A nondominated point has no competitor that is at least as accurate and no more costly with one strict improvement. We also implement a HAL-inspired origin-anchored convex-hull reconstruction and a 5% tolerance sensitivity where dominance requires 5% lower cost and 5% higher accuracy. The tolerance is a near-tie stress test, not a calibrated measurement-error model.

4 Results

4.1 Raw cost-rank transfer is explained by stable label cost propensity

The robust raw correlations invite a confound check: cost rankings may encode persistent displayed-label price/cost tiers rather than portable execution dynamics. The frozen CSV lacks token components, provider alias, and historical price-card fields, so we cannot claim a list-price adjustment. We fit an observational pooled log-cost model on 86 Generalist rows with benchmark and displayed-label effects. It explains 82.0 percent of log-cost variance, with adjusted R-squared of 72.7 percent. The six high-overlap raw correlations collapse after residualization: none of the residual-cost rank correlations is significant under a 20,000-draw permutation test. Thus raw cost transfer largely reflects stable observed label-level cost propensity. This propensity may combine provider price tier, reasoning configuration, token use, cache treatment, routing, and unrecorded setup; it is not identifiable as list price from the public CSV. Exact adjusted coefficients are reported in the anonymous supplement. The raw cross-workload rank-transfer matrix, including each pair’s Spearman correlation and shared-label count, is provided in the anonymous supplement. The raw dollar-cost structure is largely absorbed by stable displayed-label cost propensity.

4.2 Frontier membership is non-universal

No label tested on at least five workloads lies on every cost–accuracy frontier. This conclusion survives nondominance, the HAL-inspired hull, and 5% near-tie tolerance. The stronger statement that no label appears on more than 4/6 frontiers does not survive: under tolerance, o4-mini Low appears on 5/6. We therefore retain only the robust non-universal-winner conclusion.

Table 1: Higher-overlap pairs. Brackets are configuration-resampling sensitivity intervals; ND is nondominated-frontier Jaccard similarity on the shared-label cohort.

Pair	N	Accuracy ρ	Cost ρ	ND Jaccard
CORE-Bench Hard–GAIA	14	0.60 [0.16, 0.81]	0.93 [0.72, 0.99]	0.43
CORE-Bench Hard–SWE-bench Mini	14	0.48 [-0.05, 0.82]	0.74 [0.29, 0.93]	0.25
CORE-Bench Hard–TAU Airline	12	0.50 [-0.12, 0.86]	0.78 [0.26, 0.98]	0.17
GAIA–SWE-bench Mini	17	0.71 [0.31, 0.91]	0.89 [0.61, 0.99]	0.33
GAIA–TAU Airline	14	0.58 [0.14, 0.85]	0.86 [0.58, 0.97]	0.29
SWE-bench Mini–TAU Airline	14	0.76 [0.40, 0.91]	0.82 [0.46, 0.97]	0.43

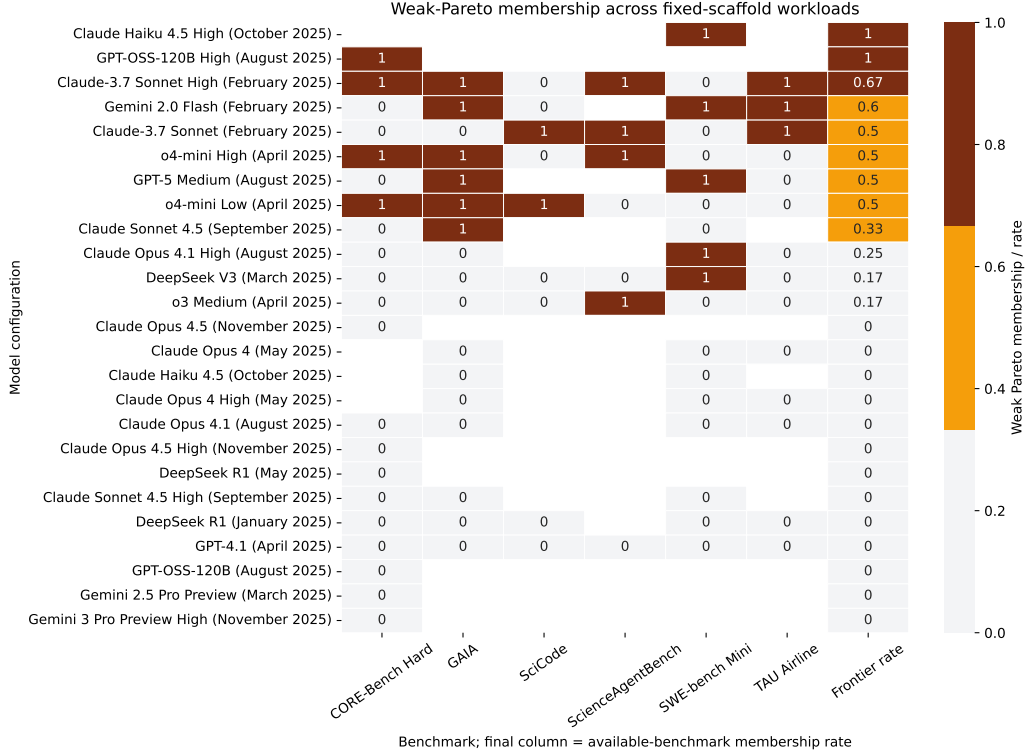


Figure 1: Nondominated cost–accuracy membership across primary workloads. The final column is each label’s rate over available workloads. No broadly tested label appears on every available frontier.

4.3 Scaffold and frontier definitions matter

The source HAL labels recover 30 rows under the hull reconstruction, with two discrepancies; ordinary nondominance includes 19 additional points. A discrete buyer choosing one system and a deployer willing to randomize between systems therefore face different decision frontiers. Within individual benchmarks, scaffolds also differ substantially: across 15 shared labels on SWE-mini, displayed SWE-Agent averages 27.4 percentage points higher accuracy than the Generalist scaffold, with higher median cost. These contrasts motivate fixing the scaffold; they are not a general causal scaffold estimate.

5 Discussion and threats to validity

A single benchmark can supply a useful prior about *raw* observed dollar cost, but the analysis does not establish portable workload-specific execution-cost dynamics: the raw cost-rank signal is largely absorbed by stable displayed-label cost propensity. Because this propensity may include provider price tier, reasoning configuration, token volume, cache treatment, routing, and unrecorded setup, it cannot be reduced to a list-price claim from the frozen public CSV. Practitioners should validate both

97 cost and token behavior on representative target tasks rather than treating one leaderboard frontier as
98 a universal efficiency mechanism.

99 Construct validity is limited because public Models and Agent Name strings do not fully encode
100 prompts, tools, budgets, harness versions, or pricing assumptions. The CSV also lacks provider
101 aliases and token components, precluding historical list-price or common-tariff reconstruction; public
102 trace archives are encrypted. Matching a display label under the Generalist scaffold is not a controlled
103 base-model comparison. Internal validity is limited by sparse, unbalanced coverage and mostly
104 single-run evaluations. DeepSeek R1 and Gemini 2.0 Flash are absent from ScienceAgentBench, but
105 the public table provides no failure or exclusion reason; missingness is unidentifiable rather than
106 assumed random. External validity is restricted to this frozen HAL snapshot and six workloads.

107 6 Conclusion

108 Raw dollar-cost ranks transfer robustly across several high-overlap public HAL workload pairs, but
109 pooled label and benchmark cost propensity explains most of this signal. The public CSV does not
110 establish portable workload-specific execution-cost dynamics or isolate historical provider list prices.
111 Cost–accuracy frontier membership is nevertheless non-universal under multiple decision-frontier
112 definitions. The resulting lesson is not that cross-benchmark cost data are useless, but that raw
113 dollar-cost leaderboards should be interpreted as a mixture of persistent label-level cost propensity
114 and workload behavior, then validated on target tasks.

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